

Write C# solutions or following programs

Problem Statement-1

Given an array of integers, calculate the **sum of all elements**.

Input

- First line contains integer n (size of array)
- Second line contains n space-separated integers

Output

Print the sum.

Example

Input:

5
1 2 3 4 5

Output:

15

Problem Statement-2

Given a string, count the number of vowels (a,e,i,o,u) present.

Input

hello world

Output

3

problem Statement-3

Find the **second largest distinct element** in an array.

Input

6
10 20 4 45 99 99

Output

problem Statement-4

Check whether a given string is a **palindrome** (same forward and backward).

Input

madam

Output

Palindrome

Problem Statement-5

Read a text file and count how many times each word appears.

Input

A file named data.txt containing:

hello world hello csharp

Output

hello: 2

world: 1

csharp: 1